Design Research Society

DRS Digital Library

DRS Biennial Conference Series

DRS2026: Edinburgh

Jun 8th, 9:00 AM - Jun 12th, 5:00 PM

Changing systems from within: Workarounds in sociotechnical system transformation

Tilda Berensson
KTH Royal Institute of Technology, Sweden

Mia Hesselgren
KTH Royal Institute of Technology, Sweden

Follow this and additional works at: https://dl.designresearchsociety.org/drs-conference-papers

 Part of the Art and Design Commons

Citation

Berensson, T., and Hesselgren, M. (2026) Changing systems from within: Workarounds in sociotechnical system transformation, in Simeone, L., Gray, C. M., Verhoeven, A., de Götzen, A., Bakırlıoğlu, Y., Zohar, H., Stead, M., and Buwert, P. (eds.), *DRS2026: Edinburgh*, 8–12 June, Edinburgh, United Kingdom. https://doi.org/10.21606/drs.2026.727

This Research Paper is brought to you for free and open access by the DRS Conference Proceedings at DRS Digital Library. It has been accepted for inclusion in DRS Biennial Conference Series by an authorized administrator of DRS Digital Library. For more information, please contact library@thedrs.org.

Changing systems from within: Workarounds in sociotechnical system transformation

Tilda Berensson*, Mia Hesselgren

[a]Department of Engineering Design, KTH Royal Institute of Technology, Sweden

*Corresponding author e-mail: tildab@kth.se

doi.org/10.21606/drs.2026.727

Abstract: Systems are shaped by dominant mindsets that constrain which futures can be imagined or made possible. Supporting system transformation requires shifting temporal boundaries and revealing the hidden system logics. This study explores how design can contribute to understanding and reframing sociotechnical system change from within. In an empirical study conducted within a research consortium investigating the electrification of road freight transport, we, as design researchers, entered the system change as participants. Using a practice-oriented lens, we learned from riding in trucks, conducting contextual observations, and interviewing drivers and transport planners whose everyday work sustains the system. What is often portrayed as a tightly controlled, planned system instead revealed itself as held together by human skills. This study uncovered how systems are maintained by *workarounds* – established practices and improvised actions that expose hidden tensions and logics. We propose that workarounds can be used as potential leverage points for transformative change.

Keywords: transformative design practices; systemic transformation; sociotechnical systems; leverage points

1. Introduction

Every day, thousands of trucks move goods across roads, sustaining the supply chains that societies depend on. However, this constant movement comes at substantial costs: the transport sector, and specifically road freight, remains a significant source of emissions, for which electrification has been suggested as a path to reductions (Aryanpur & Rogan, 2024; Niu, 2025). Transforming this system, however, is far more complex than replacing engines or fuels. As Kania et al. (2018) argue, systemic change requires shifting the conditions that sustain the problem. In the case of electrifying road freight, this involves reshaping practices, regulations, decision-making, investment priorities, infrastructure, and the underlying logics through which the system is understood and maintained (Raoofi et al., 2025). The visibility of these systemic conditions varies within the system: some, such as regulations and routines,

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International Licence.

are formal and measurable, while others, such as underlying logics, are tacit and lived, making some conditions easier to target than others (Kania et al., 2018).

Current attention to the electrification of road freight tends to focus on what is most visible and measurable: developing new policies, vehicles, batteries, planning routines, and optimisation models (Gillström et al., 2024). While necessary, such interventions risk leaving deeper conditions untouched, including the assumptions, relations, and everyday tensions that sustain the current system. Moreover, when futures are imagined primarily through extrapolation from present knowledge and practices, the assumptions we carry today become the boundaries of what we can envision (Robinson, 1988). The result is a risk of continuously reproducing current trajectories, even in the pursuit of change.

This raises a critical question for us as design researchers aiming to support systemic transformation: how can we work with the present system and its transformation without reproducing its logics in the future? In this paper, we begin to explore this question by attending to what actors within the system are already doing. Drawing on Hummels et al. (2019), we understand transformation as an irreversible, long-lasting shift in values, worldviews, and both technical and social arrangements. According to them, design is positioned as a practice that operates from within such processes, alongside those expected to enact it, helping people, communities, or societies recognise the need for change, develop new ways of working toward it, and embody it in everyday practice. This understanding of the designer's role and the broader ambition to support system actors in engaging in transformative practices inform the approach taken in this paper.

As part of a research consortium working toward the electrification of long-distance road freight, we conducted an empirical study with truck drivers and transport planners. Through close engagement with the everyday work of these practitioners, recurring adjustments and improvisations became visible across logistics flows. We conceptualise these recurring patterns as *workarounds*, through which system logics, relations, and tensions can become visible. Building on Kania et al.'s (2018) notion of shifting the conditions that hold problems in place, we propose that workarounds can signal potential *leverage points* for systemic transformation (Meadows, 1999). Rather than prescribing change from outside, attending to workarounds offers a practical entry point, one that begins with the system's own signals of what needs to shift. In doing so, the paper contributes with an empirically grounded approach to identifying potential leverage points within ongoing system transformations by conceptualising workarounds as indicators of systemic tension.

This article is structured as follows: First, we frame road freight as a sociotechnical system, introduce emergence and workarounds as key concepts, and position design practice in relation to leverage points and transformative engagement. Second, we describe the empirical study through which recurring adjustments were identified and conceptualised as workarounds. Finally, we reflect on these findings as transformative leverage points and discuss implications for transformative design practice.

2. Theoretical framing

While the electrification of road freight is often described as a sociotechnical *transition*, we use the term *transformation*. Where transition emphasises movement toward a new system configuration, transformation draws attention to the ongoing process through which such change takes shape, the shifting of values, relations, and practices that make it possible (Hummels et al., 2019; Shove & Walker, 2007). However, we draw on sustainability transitions research to understand how such processes of reconfiguration unfold within established sociotechnical systems, and to situate the electrification of road freight within this broader dynamic.

2.1 Transforming a sociotechnical system

Road freight transport is a sociotechnical system, one in which technical artefacts, human practices, and institutional structures are deeply entangled (Vermaas et al., 2022). What makes such systems challenging to transform is not any single component, but the interdependence among them. Introducing electric trucks and charging infrastructure requires renegotiating roles, responsibilities, and the logics that organise everyday operations. However, established practices and institutional norms hold current configurations in place, stabilising what is familiar and resisting the opposite. No single actor can transform such a system on their own. Consequently, collaboration across organisational and disciplinary boundaries becomes essential (Frantzeskaki et al., 2012).

A sociotechnical perspective means attending not only to technological change but also to the relations, practices, and assumptions that sustain the present system. For design research to support systems transformation, we must first understand how and where it becomes perceptible in practice.

2.2 Emergence at the margins of practice

Sustainability transitions research suggests that change is preferably understood as a gradual reconfiguration within existing sociotechnical systems, rather than as the linear replacement of one system by another (Geels, 2011; Loorbach et al., 2017). Within these processes, new configurations often emerge at the margins of established systems (Laurell Thorslund et al., 2025), growing within the structures they later come to challenge. The Two Loops Model (Wheatley & Frieze, 2006), as further developed by Laurell Thorslund et al. (2025), illustrates this as an overlap between decline and emergence: as dominant systems weaken, alternatives develop in parallel, nourished by new practices, values, and relations. Therefore, as sociotechnical systems begin to reconfigure, tensions might emerge between established practices and the logics of emerging alternatives.

We take the perspective that the emergence of new practices and logics becomes visible when established routines no longer reproduce themselves seamlessly. Following Heidegger's account of *breakdown* (Koschmann et al., 1998), systems remain largely invisible while functioning smoothly; it is when practices are interrupted that underlying structures, such as relations, dependencies and values, become explicit and perceptible. The concept of *workarounds* (Alter, 2014) offers a complementary lens. Workarounds occur when established procedures diverge from actual conditions, prompting adaptations and

improvisations that maintain system functionality. While not transformative in themselves, they expose tensions in existing systems, revealing where boundaries and routines hinder desired actions. From a systems change perspective, these situated responses illuminate the contours of emerging possibilities within the present system.

Practice-oriented perspectives deepen this understanding by locating change in the everyday practices through which systems are sustained and contested (Shove & Walker, 2010). Here, workarounds are not merely deviations but situated adaptations through which transformation becomes visible in practice, enacted by those who inhabit the system.

2.3 Design practice in systems transformation

If workarounds reveal where transformation already unfolds within a system, the question becomes how design can engage with these expressions of systemic tension.

Within systems theories, *leverage points* offer a lens for where such engagement might catalyse change. Meadows (1999) conceptualises these as places within a system where small shifts can trigger significant systemic effects. Among these, shifts in mental models, which refer to the underlying beliefs and assumptions that shape system behaviour, are considered particularly transformative (Meadows, 1999; Van Der Bijl-Brouwer et al., 2024). Similarly, Kania et al. (2018) argue that sustained change requires attention to deep structures of thought, not only explicit rules. Recent work further emphasises the need for practical methods to help designers identify and address such points (Laven et al., 2025).

However, engaging with leverage points is not a neutral act. Transitions increasingly unfold in interdisciplinary constellations in which researchers, companies, and public institutions collaborate toward shared goals, and different epistemic cultures meet (Knorr Cetina, 2007; Robinson, 2008). Within such contexts, framing becomes a critical act (Dorst, 2015), shaping what is made visible, whose perspectives count, and which futures are made possible – decisions that are entangled with power (Schmidt et al., 2020; Turnhout et al., 2020). Design is increasingly invited into these settings, yet its role often remains ambiguous. Therefore, the question is not only what designers do in these constellations but how they position themselves in relation to the system they seek to change (Irwin et al., 2020).

Reframing design-as-practice (Kimbell, 2009) shifts attention from the design of artefacts to situated actions and relations, positioning designers as participants whose knowledge emerges through engagement. Scott et al. (2012) demonstrate that designing for sustainability requires enabling the reconfiguration of existing norms and ways of doing, rather than delivering solutions. This also entails attending to what is under strain. As Shove and Walker (2007) explain, transitions involve not only emergence but also displacement and loss. Supporting transformation from within, therefore, presupposes that designers enter the system, understand ongoing practices, and build relational trust with those who inhabit it. Such engagement requires a reflexive practice, in which the assumptions, dependencies, and values that guide collective action are made visible (Hummels et al., 2019; Irwin, 2015; Schmidt et al., 2020).

From this situated and reflexive positioning of design, we argue that workarounds become relevant not only as analytical objects but also as design material. When approached as materials for dialogue and collective learning, they offer the potential to make visible how

everyday actions both sustain and contest the underlying logics of the system, providing concrete entry points into the mindset-level shifts that transformative change requires.

To explore how such systemic tensions become visible in practice, the following presents an empirical study conducted within the Swedish road freight system, where the introduction of electric trucks is creating friction between established and emerging logics. We return to the proposition of workarounds as transformative leverage points in the discussion.

3. Learning from within

This section presents the empirical study conducted within a large-scale system demonstration consortium aimed at advancing the electrification of long-distance road freight transport in Sweden. The consortium brings together researchers from multiple disciplines alongside industry partners, including truck manufacturers, hauliers, energy providers, and charging operators. As design researchers in this context, we aim to contribute knowledge about the human and relational dimensions of this transformation, emphasising that systems are created and maintained by people who are skilled in their respective fields. Just and sustainable transformations need to recognise these capacities (Gong et al., 2025).

The study investigated two groups of practitioners: four truck drivers and four transport planners, each responsible for one of four logistics flows within the consortium. By accompanying their daily work, we aimed to understand where tensions, resistance, and inertia emerge in the shift toward electric freight. The sections below outline the research stance, data collection, and analytical process that led us to focus on workarounds as signals of systemic tension.

3.1 Research stance

We approached the study with a *practice-oriented stance*, meaning that practices, rather than individuals, served as the primary unit of analysis (Schatzki, 2005; Shove, 2010). Rather than asking drivers and transport planners what they *think* about electrification, we were interested in what they *do*: the routines, skills, and material arrangements through which the freight system is sustained day to day (Shove et al., 2012).

Informed by Hummels et al.'s (2019) understanding of design as a practice that works from within transformation, we adopted a first-person perspective and treated our situated engagement with practitioners as a form of inquiry. Rather than observing change from a distance, we entered the system to learn about electrification and its impacts on road freight alongside those who sustain it, a step we consider valuable before contributing to its transformation through design interventions.

Entering someone else's everyday context is, however, never neutral. As researchers, we inevitably bring our own perspectives into the field, and within some professional groups, researchers are not automatically trusted with the real stories of daily work. Therefore, it was important that the study was designed to signal genuine interest in drivers and transport planners as skilled professionals, and to foster a collaborative rather than an extractive approach.

3.2 Methods

The study was conducted in two phases, designed to progressively build understanding across the four logistics flows. The first phase focused on engaging with truck drivers to understand how the freight system operates day to day. The second phase consisted of interviews with transport planners responsible for the same logistics flows. Adapting to practitioners' working routines was an important consideration, allowing close engagement while minimising disruption to their work.

3.2.1 Phase 1: Truck drivers

Truck drivers are the actors who carry the system every day and are directly affected by the transition to electric trucks. Despite this, their perspectives are often overlooked in discussions about the system transformation. Therefore, we chose to begin by studying the road freight system from the drivers' perspective.

Preparing for the in-person visits, we introduced a *sensitising material* (Sanders & Stappers, 2012) to shift attention away from technological discussions of electrification toward drivers' own experiences and values. Hence, each truck driver was asked to complete a short diary over five days prior to the field visits. The diaries served two purposes: to encourage reflection in a familiar setting and to begin establishing a relationship before meeting in person. Questions invited drivers to reflect on what brought them to the profession, what they take pride in, and how their daily lives are shaped by their work. These reflections provided initial starting points for the conversation during the field visits.

After the diaries were completed, each driver was accompanied during their daily route for six to ten hours. This co-riding phase combined contextual observation with open-ended conversations, inspired by Czarniawska's (2014) understanding of fieldwork. Being present in the truck allowed access to tacit, embodied aspects of practices that are difficult to reach through interviews alone. Conversations were guided by curiosity rather than a fixed interview guide, although prepared questions were available as support.

3.2.2 Phase 2: Transport planners

Following the field engagement with truck drivers, semi-structured interviews were conducted online with the four transport planners responsible for the same logistics flows. Their coordinating role with drivers is central to how the freight system functions and, therefore, to how electrification reshapes it. Conducting these after the field visits allowed questions to be grounded in observations of the drivers' work. It also signalled that efforts had been made to understand the practical realities of road freight transport. The interviews focused on transport planners' roles, responsibilities, and their experiences of electrification. Follow-up questions explored how scheduling decisions were made and whether drivers' tacit knowledge was recognised in planning processes.

3.3 Analysis

The empirical material, in the form of fieldnotes, recordings and observations, was collected and analysed by the first author, with emerging interpretations discussed iteratively with the co-author throughout the process.

The analysis was inductive, beginning without predefined categories. Guided by the same practice-oriented lens that informed the fieldwork (Schatzki, 2005; Shove, 2010; Shove et al., 2012), attention was directed not only to what drivers and transport planners said, but also to what they did, including how tasks were carried out, how stressful situations were handled and how materials and infrastructures shaped everyday work. Following Czarniawska's (2014) suggestions, fieldwork was understood as producing material shaped in the movement between field and desk. After each field visit, notes and recordings were reviewed and organised by clustering insights and quotes thematically through visual mapping in Miro, enabling comparison across the four logistics flows as the study progressed.

In the first co-riding and interviews, certain adjustments and apparent rule-bending in daily work were noted. When similar situations appeared in subsequent flows, they began to look less like individual choices and more like structural responses to recurring constraints. Moving back and forth between field visits and desk reflections, it became clear that drivers and transport planners were adjusting to or departing from formal procedures in similar ways, for similar reasons.

Engaging with existing literature led us to conceptualise these patterns as *workarounds* (Alter, 2014), understood not as errors or rule-breaking, but as socially learned responses to the gap between the system's intended functions and its actual operations. Once conceptualised, workarounds became an analytical lens for re-reading the material, surfacing tensions between system logics and lived routines. The following section presents the empirical findings based on this analysis.

4. A system held together by human skills

The road freight system is often portrayed as a tightly coordinated, optimisable network of logistics and schedules, designed to match societal demands with the steady movement of goods. It appears measurable and optimisable, a machine of predictable flows. However, from both the truck driver's seat and the transport planner's desk, our study reveals a different picture: beneath the planned surface lies a reality of obstacles, disruptions, and everyday negotiations. These findings directly reflect systemic conditions, ranging from explicit rules to implicit logics, and demonstrate how they are played out in everyday work.

In this section, we present the workarounds surfaced in the analysis and examine what they reveal about the system's underlying conditions at three levels of visibility. First, on the explicit level, the identified workarounds highlight where formal procedures no longer align with everyday realities. Second, on the semi-explicit level, they illuminate who carries friction and who has agency. Third, on the implicit level, the workarounds make the mental models that stabilise the current system visible, thereby indicating where transformative change would be required.

4.1 Explicit: Navigating rules and realities

At the most visible level, workarounds emerged where formal rules, schedules, and tools failed to align with the realities of daily work. We found in the contextual observations that drivers repeatedly had to choose between following regulations and maintaining the flow of

goods. As one driver explained, when negotiating between customer bookings and load regulations:

> "If I start backing in and reloading in the morning, my boss calls and says, 'What the hell, you're supposed to be there at 9,' so it gets kind of sensitive there too."

These tensions appeared not only in logistic operations but also when driving on the road. Because road layouts, parking spaces, and loading areas are rarely designed for heavy trucks, drivers often manoeuvre in ways that technically breach traffic rules, for instance, briefly driving on the wrong side or stopping in prohibited areas.

Two particularly common workarounds identified in the study illustrate how such negotiations have become normalised among drivers. The first is the practice of dropping the trailer in a nearby location to collect later. Through shared experience and social learning, drivers know which informal parking spots are accepted and plan their routes and loading accordingly. This workaround of space usage was observed during all co-rides where a trailer was attached to the truck.

The second workaround concerns the usage of time and was expressed with several examples. Break and drive times are tightly regulated; however, delivery windows do not allow for everyday unpredictability. Therefore, in practice, following the rules risks delays, making many drivers prioritise completing essential tasks first and log "break" time while the truck is stationary during loading or unloading. This protects them from fines and creates flexibility to take a real break later, when it fits the actual rhythm of the day rather than the timetable or legislation. As portrayed in this example:

> "This schedule I have even tells me when to take my break, but it's not exactly fun if my long break, which is supposed to be for meals, ends up being at ten o'clock. But that's what a program has said then, but there's a gap between the program and reality."

This bending of the break rule was far from hidden, as another driver admitted:

> "Any driver who says they don't do it is lying."

4.2 Semi-explicit: Drivers as system buffers

Beyond formal rules and procedures, the workarounds revealed how friction is absorbed between people. In practice, drivers become system buffers, absorbing the impact of delays or tight schedules to maintain the system's functionality. By shortening or skipping breaks, establishing direct contact with customers, or finding creative ways to unload goods, drivers explained how they regularly prevent single disruptions from causing further delays. As one driver described:

> "It pretty much happens all the time that something goes wrong, waiting times and such, and then you just have to find a way to work around it so you don't lose time."

Another driver explained:

> "I come here often, so I know XX who works there, and if there's a queue and I'm a bit late, he doesn't mind."

This buffering role relies on local, situated knowledge. When drivers deliver to the same places, they develop an intuitive understanding of routes, sites, people, and recurring frictions. Such tacit knowledge rarely appears in digital planning systems, despite planners and, ultimately, the system itself depending on it. As one planner noted,

> "As the one only planning the trucks, what I actually see is just a registration number and a route."

An illustrative example of how drivers increasingly become system buffers in the emerging electric system was found when one driver described how he uses his weekly day off to move the truck from the charging station to the terminal, ensuring a smooth start the following day. Without this personal adjustment and individual workaround, which he claimed his manager was unaware of, he risked falling behind schedule. In this sense, the driver's spare time became a buffer, compensating for new frictions that appeared with the electrification of this logistics flow.

4.3 Implicit: Persisting diesel logics

The same workarounds also point to changes needed at a less visible layer: the persistence of "diesel logics" in an emerging electric system. At this deepest level, workarounds illuminate where the system itself begins to challenge the logics that currently sustain it. We found that in planners' everyday decisions, the values of speed, efficiency, and customer satisfaction continued to guide how routes were scheduled, delays were handled, and what was considered acceptable use of time and capacity. In interviews, planners emphasised that the customer always comes first, even when this led to half-full trucks or scattered deliveries, as one planner noted:

> "Well, we need to get the goods out no matter how much or little there is"

Drivers also described this inefficiency:

> "Can't we hold off on that delivery since it's such a small volume today, given that we're delivering on Wednesday anyway?"

Despite this awareness, company policies and customer contracts were often portrayed as rarely allowing for change possibilities, and the logic of cheapest wins continued to dictate operations, leaving little space for shared logistics or adaptive planning. Electrification, with its need for coordinated charging, flexible timing, and shared infrastructure, was described as colliding with a system optimised for fossil-fuel efficiency. As one driver explained:"No company wants to pay extra for the driver to stand still and charge."

Building further on logics that do not match realities, organisational instructions assume that logistic flows can be completed seamlessly. In reality, logistics performance depends on the individuals who are carriers of the system. All four planners underlined the importance of planning for people, not just vehicles, despite instructions not to do so from above, indicating that changes are needed in terms of what is considered normal:

> "When you work with the same drivers, you know who takes more time and who's quicker. You always plan based on who's driving, which truck it is, and which stops they have."

Taken together, these empirical findings reveal how workarounds expose systemic tensions at various levels of visibility, which we now examine as a basis for our reflective discussion of workarounds as transformative leverage points.

5. Workarounds as transformative leverage points

In the introduction, we asked how design researchers can engage with a system in transformation without reproducing its existing logics. The workarounds surfaced in this study offer a grounded, if partial, response. They were not something we set out to find; they emerged through close engagement with practitioners whose everyday work sustains the road freight system. It was through analysing these recurring patterns, such as truck drivers logging breaks while loading, transport planners relying on tacit knowledge, both bridging the gaps between rules and realities, that workarounds became a meaningful analytical concept. In what follows, we reflect on what this concept can offer design research engaging with systemic transformation.

Each workaround observed exposes a meeting point between formal rules and established procedures and the requirements of lived practices. It is at these intersections that the system's underlying logics, in our case, its values of efficiency, punctuality, optimisation algorithms, and customer primacy, are both upheld and contested. Following Heidegger's account of breakdown (Koschmann et al., 1998), these are moments when the system becomes illuminated: what normally operates invisibly is made perceptible through disruption. As Hesselgren (2019) suggests through the notion of *glitches* as design material, such ruptures are not problems to be solved but signals to be read.

This suggests that workarounds, when treated as analytical tools rather than operational problems, can indicate where change is required. By surfacing and reflecting on workarounds together with those inside the system, designers can help make explicit the implicit mental models that shape behaviour and decision-making. This means working with the system’s own signs of transformation to explore what might need to be reconfigured, rather than prescribing change from the outside.

As Laurell Thorslund et al. (2025) argue, allowing a dominant system to decline while a new one emerges involves deciding what to preserve, challenge, or let go. Hummels et al. (2019) similarly emphasise that transformative practices require engaging with present practices and enabling people to reflect on how their everyday actions sustain or contest existing logics. By treating workarounds as design material, designers can provoke such reflections among actors: which logics sustain the current system, and which ones must be left behind to allow something new to grow?

Engaging with workarounds may involve creating spaces for system actors to collectively reflect on the tensions revealed. Questions such as why these adjustments are necessary, what values they expose, and what might change if these logics were challenged can open possibilities for reimagining the system.

This study thereby responds to calls within design research for practical ways of identifying where and how designers can intervene in systemic transformations (Laven et al., 2025; Van Der Bijl-Brouwer et al., 2024). By conceptualising workarounds as situated indicators of

systemic tensions, the paper contributes with an empirically grounded approach for recognising potential leverage points within ongoing transformations.

6. Implications for transformative design practices

In this study, we explored how recurring workarounds in everyday logistics practices revealed systemic tensions as road freight undergoes electrification. The findings aim to build further on the situated positioning of the designer (Hummels et al., 2019; Kimbell, 2009; Scott et al., 2012). In the following, we discuss the implications for design research engaging with systemic transformation in terms of entering and reading the system, the ethics of working with workarounds, and the study's positioning within broader transition research.

6.1 Being in the system

Being in the system and learning through people and practice proved essential for understanding how it functions. Attending to everyday tensions and frictions revealed how structures and logics play out, insights that could not have been reached through consortium meetings or transport conferences alone. When engaging with sociotechnical transitions, such as the shift toward electric freight, it is crucial to stay curious about the present system, as it forms the foundation for imagining what comes next.

We therefore argue that such situated presence is not merely a methodological choice but a prerequisite for design engaging with systemic transformation. The diary exercise prompted initial reflection among drivers, but it was during the long hours in the trucks that previously tacit actions became visible and were transformed into shared points of awareness for both researchers and drivers. This illustrates how “being in the system” enables designers to identify not just what people do, but where they already adjust and negotiate the system from within.

However, being in the system is not sufficient on its own. Questioning ingrained ways of working and the values that guide change can easily be perceived as criticism, yet transformation requires doing just that, which calls for a reflexive posture (Hummels et al., 2019; Irwin, 2015). As researchers, our assumptions inevitably shape what we attend to and how we interpret what we see, thereby influencing what becomes visible. In this study, focusing on workarounds in the analysis, rather than on presumed answers, helped maintain an open, situated view, allowing us to learn with people rather than merely about them.

Workarounds can also help bridge the gap between researchers and system actors. Because they emerge from actors’ own practices, they can potentially be easily engaged with. Once surfaced, they make shared logics visible without assigning blame, offer recognition, and open space for reflection and dialogue. Simultaneously, they suggest that not everything needs to change. Some reveal practices and knowledge worth preserving, while others signal logics that may need to be challenged. Shove and Walker (2007) argue that transitions involve not only emergence but also displacement and loss. Hence, we suggest workarounds, conceptualised as signals of systemic tension, offer a way of attending to both.

Based on these reflections, we propose that design research engaging with systemic transformation could involve three interconnected steps. The first is *entering* the system and building relational trust through situated engagement with those who inhabit it. The second is *reading* it by identifying where established routines are already being adjusted or negotiated, treating workarounds as expressions of systemic tension rather than as problems to be fixed. The third is *engaging* system actors through the workarounds identified, using them as materials for collective reflections on which logics, rules, or relations to preserve, challenge, or leave behind. This study has demonstrated the first two steps, while the third remains the focus of ongoing work within the research consortium.

6.2 Workarounds as an act of care

By treating workarounds as design materials, transformative designers can initiate dialogue from lived examples rather than abstract ideas, helping shift the designer's position from an external critic to a partner articulating what is already happening in the system. Surfacing workarounds, however, requires care. If addressed without reflection, they risk being used to justify further automation, tighter control, or additional demands on those who already carry the system, rather than opening space for shared responsibility and learning. Transformation inevitably involves questions of power: who defines the problem, whose work is recognised, and who bears the consequences of change. Such questions require careful consideration by transformative designers.

Therefore, to intervene in a system is also an ethical act. It is difficult to foresee how changes in what is challenged, preserved, or left behind will affect those within it. Working with workarounds as transformative leverage points should thus be done collaboratively, not as a way of fixing people's behaviour, but as an invitation to reflect on and collectively reimagine what the emerging system could become. In this sense, using workarounds as design material can be understood as an act of care: supporting actors in recognising their role in sustaining or changing system logics, and enabling transformation to emerge from within the system rather than being imposed upon it.

6.3 Situating the contribution

This study focuses on transformation as an ongoing process and draws on sustainability transitions research to understand how reconfigurations unfold within established systems. While transition research has examined the system-level dynamics of transformation, less attention has been paid to how actors navigate these shifts in everyday practice (Shove & Walker, 2010). Recent work has called for greater attention to the tensions that emerge as actors navigate transformation, including research on which actors matter, what roles they take, and how they relate to the system (Gillström, 2024; Hyysalo et al., 2022). The workarounds surfaced in this study offer one empirical entry point for exploring such questions.

The study is situated in Sweden and follows four logistics providers that are already engaging in the electrified system. We acknowledge that this shapes the scope of our findings. Further research is needed to investigate whether similar patterns emerge in other contexts, and particularly in settings where electrification has not yet begun.

This paper shows that workarounds, when approached through a practice-oriented lens, can reveal systemic tensions across levels of visibility and signal where mindset-level change is needed. By entering the system and learning alongside its actors, design researchers can identify these tensions as potential leverage points, grounding transformative design in the practices of those who sustain the system.

7. References

Alter, S. (2014). Theory of Workarounds. *Communications of the Association for Information Systems*, *34*. https://doi.org/10.17705/1CAIS.03455

Aryanpur, V., & Rogan, F. (2024). Decarbonising road freight transport: The role of zero-emission trucks and intangible costs. *Scientific Reports*, *14*(1), 2113. https://doi.org/10.1038/s41598-024-52682-4

Czarniawska, B. (2014). *Ute på fältet, inne vid skrivbordet: 1:1*. Studentlitteratur AB.

Dorst, K. (2015). Frame Creation and Design in the Expanded Field. *She Ji: The Journal of Design, Economics, and Innovation*, *1*(1), 22–33. https://doi.org/10.1016/j.sheji.2015.07.003

Frantzeskaki, N., Loorbach, D., & Meadowcroft, J. (2012). Governing societal transitions to sustainability. *International Journal of Sustainable Development*, *15*(1–2), 19–36. https://doi.org/10.1504/IJSD.2012.044032

Geels, F. W. (2011). The multi-level perspective on sustainability transitions: Responses to seven criticisms. *Environmental Innovation and Societal Transitions*, *1*(1), 24–40. https://doi.org/10.1016/j.eist.2011.02.002

Gillström, H. (2024). Barriers and enablers: How logistics companies could tackle the transition to electrified road freight transport. *Cleaner Logistics and Supply Chain*, *13*, 100172. https://doi.org/10.1016/j.clscn.2024.100172

Gillström, H., Jobrant, M., & Sallnäs, U. (2024). Towards building an understanding of electrification of logistics systems – A literature review and a research agenda. *Cleaner Logistics and Supply Chain*, *10*, 100134. https://doi.org/10.1016/j.clscn.2023.100134

Gong, J., Xylia, M., Strambo, C., Nykvist, B., & Celik, S. (2025). What happened to the driver? Implications of electrification, digitalization, and automation on truck and taxi drivers. *Technology in Society*, *81*, 102816. https://doi.org/10.1016/j.techsoc.2025.102816

Hesselgren, M. (2019). *Designing for sustainability practices: RE-DO Design Doings, Strategies and Postures* [Doctoral dissertation]. KTH Royal Institute of Technology.

Hummels, C. C. M., Trotto, A., Peeters, J. P. A., Levy, P., Alves Lino, J., & Klooster, S. (2019). Design research and innovation framework for transformative practices. In *Strategy for change* (pp. 52–76). Glasgow Caledonian University.

Hyysalo, S., Heiskanen, E., Lukkarinen, J., Matschoss, K., Jalas, M., Kivimaa, P., Juntunen, J. K., Moilanen, F., Murto, P., & Primmer, E. (2022). Market intermediation and its embeddedness – Lessons from the Finnish energy transition. *Environmental Innovation and Societal Transitions*, *42*, 184–200. https://doi.org/10.1016/j.eist.2021.12.004

Irwin, T. (2015). Transition Design: A Proposal for a New Area of Design Practice, Study, and Research. *Design and Culture*, *7*(2), 229–246. https://doi.org/10.1080/17547075.2015.1051829

Irwin, T., Tonkinwise, C., & Kossoff, G. (2020). Transition Design: An Educational Framework for Advancing the Study and Design of Sustainable Transitions. *Cuadernos Del Centro de Estudios de Diseño y Comunicación*, (105). https://doi.org/10.18682/cdc.vi105.4188

Kania, J., Kramer, M., & Senge, P. (2018). *THE WATER OF SYSTEMS CHANGE*. FSG.

Kimbell, L. (2009). *Beyond design thinking: Design-as-practice and designs-in-practice*. CRESC Conference.

Knorr Cetina, K. (2007). Culture in global knowledge societies: Knowledge cultures and epistemic cultures. *Interdisciplinary Science Reviews*, *32*(4), 361–375. https://doi.org/10.1179/030801807X163571

Koschmann, T., Kuutti, K., & Hickman, L. (1998). The Concept of Breakdown in Heidegger, Leont'ev, and Dewey and Its Implications for Education. *Mind, Culture, and Activity*, *5*(1), 25–41. https://doi.org/10.1207/s15327884mca0501_3

Laurell Thorslund, M., Bates, O., Eriksson, E., Pargman, D., Biørn-Hansen, A., Bakhshoudeh, F., & Menon, A. R. (2025). Meta-crisis computing and you: Finding agency through the Two Loops model of change. *Proceedings of the Sixth Decennial Aarhus Conference: Computing X Crisis*, 127–139. https://doi.org/10.1145/3744169.3744178

Laven, T., Mazé, R., & Wallace, N. (2025). *A Tale of Two Conceptual Tools: Analysing Leverage Point Tools in Relation to a Water Utility's Circular Economy Transition*. RSD14: ARCS OF IMPACT.

Loorbach, D., Frantzeskaki, N., & Avelino, F. (2017). Sustainability Transitions Research: Transforming Science and Practice for Societal Change. *Annual Review of Environment and Resources*, *42*(Volume 42, 2017), 599–626. https://doi.org/10.1146/annurev-environ-102014-021340

Meadows, D. (1999). *Leverage Points: Places to Intervene in a System*. https://donellameadows.org/wp-content/userfiles/Leverage_Points.pdf

Niu, T. (2025, October 8). Beyond trucks: Toward a greener global freight transportation system. *International Council on Clean Transportation*. https://theicct.org/toward-a-greener-global-freight-transportation-system-oct25/

Raoofi, Z., Andruetto, C., & MacDonald, R. (2025). System-level impact of electrification on road freight transport efficiency. *Npj Sustainable Mobility and Transport*, *2*(1), 42. https://doi.org/10.1038/s44333-025-00061-5

Robinson, J. (2008). Being undisciplined: Transgressions and intersections in academia and beyond. *Futures*, *40*(1), 70–86. https://doi.org/10.1016/j.futures.2007.06.007

Robinson, J. B. (1988). Unlearning and backcasting: Rethinking some of the questions we ask about the future. *Technological Forecasting and Social Change*, *33*(4), 325–338. https://doi.org/10.1016/0040-1625(88)90029-7

Sanders, E., & Stappers, P. (2012). *Convivial toolbox: Generative research for the front end of design*. BIS Publishers.

Schatzki, T. R. (Ed.). (2005). *The practice turn in contemporary theory* (Online-Ausg). Taylor and Francis.

Schmidt, L., Falk, T., Siegmund-Schultze, M., & Spangenberg, J. H. (2020). The Objectives of Stakeholder Involvement in Transdisciplinary Research. A Conceptual Framework for a Reflective and Reflexive Practise. *Ecological Economics*, *176*, 106751. https://doi.org/10.1016/j.ecolecon.2020.106751

Scott, K., Bakker, C., & Quist, J. (2012). Designing change by living change. *Design Studies*, *33*(3), 279–297. https://doi.org/10.1016/j.destud.2011.08.002

Shove, E. (2010). Beyond the ABC: Climate Change Policy and Theories of Social Change. *Environment and Planning A: Economy and Space*, *42*(6), 1273–1285. https://doi.org/10.1068/a42282

Shove, E., Pantzar, M., & Watson, M. (2012). *The Dynamics of Social Practice: Everyday Life and How It Changes*. SAGE Publications, Limited. http://ebookcentral.proquest.com/lib/kth/detail.action?docID=880778

Shove, E., & Walker, G. (2007). Caution! Transitions Ahead: Politics, Practice, and Sustainable Transition Management. *Environment and Planning A: Economy and Space*, *39*(4), 763–770. https://doi.org/10.1068/a39310

Shove, E., & Walker, G. (2010). Governing Transitions in the Sustainability of Everyday Life. *Research Policy*, *39*, 471–476. https://doi.org/10.1016/j.respol.2010.01.019

Turnhout, E., Metze, T., Wyborn, C., Klenk, N., & Louder, E. (2020). The politics of co-production: Participation, power, and transformation. *Current Opinion in Environmental Sustainability*, *42*, 15–21. https://doi.org/10.1016/j.cosust.2019.11.009

Van Der Bijl-Brouwer, M., Celik, S., De Koning, J., Nieuwborg, A., & Tromp, N. (2024, June 23). *Systemic Design Reasoning for Societal Transitions*. DRS2024: Boston. https://doi.org/10.21606/drs.2024.585

Vermaas, P., Kroes, P., Van De Poel, I., Franssen, M., & Houkes, W. (2022). *A Philosophy of Technology: From Technical Artefacts to Sociotechnical Systems*. Springer International Publishing. https://doi.org/10.1007/978-3-031-79971-6

Wheatley, M., & Frieze, D. (2006). *Using Emergence to Take Social Innovation to Scale*. The Berkana Institute.

About the Authors:

Tilda Berensson is a PhD candidate at KTH Royal Institute of Technology, Stockholm. Through a practice-oriented perspective, her research examines how transformation is enacted by those working within dominant systems and how design can support such processes.

Mia Hesselgren is an assistant professor in industrial design engineering at KTH Royal Institute of Technology, Stockholm. As a transdisciplinary design researcher, she uses practice-oriented approaches, and her research involves interventions in everyday life through designed systems that reconfigure practices.